Introduction to Operations Research, Deterministic Models

Lecture 15: Location Problems and Vehicle Routing Problems

Department of Industrial Engineering, Tsinghua University

Instructor: Junyi Liu

Dec 24th, 2024

Recall: Set Covering Problem vs Maximum Covering problem

Set Covering Problem (100 % coverage) :

$$\begin{aligned} \text{minimize} \quad & \sum_{j \in J} f_j x_j \\ \text{subject to} \quad & \sum_{j \in J} a_{ij} x_j \geq 1, \forall i \in I \\ & x_j \in \{0,1\}, \forall j \in J \end{aligned}$$

Maximum Covering problem :

$$\begin{aligned} \text{maximize} \quad & \sum_{i \in I} h_i z_i \\ \text{subject to} \quad & \sum_{j \in J} a_{ij} x_j \geq z_i, \forall i \in I \\ & \sum_{j \in J} x_j \leq P, \\ & x_j \in \{0,1\}, z_i \in \{0,1\}, \forall j \in J, \forall i \in I \end{aligned}$$

Covering-based Models

- **Set covering:** min # sites needed to cover all demand nodes with C coverage distance
- **Max covering:** max # covered demands with P sites
- **P-center:** min # coverage distance needed to cover all demand nodes with P sites

P-center problem

Problem Description

- network with node set N and arc set A: $G = (N, A)$
- set of demand nodes to be covered: $I \subset N$
- set of candidate facility sites: $J \subset N$
- maximum number of facilities: P
- distance of arc $(i, j) \in A$: d_{ij}
- ~~demand at node i: h_i~~
- ~~cost to locate a facility at candidate site j: f_j for $j \in J$~~
- ~~coverage distance: C~~

Goal: Locate a set of facilities that minimize the maximum distance between a demand node and its closest facility,

which is equivalent to minimize the coverage distance needed to cover all demand nodes with P sites

P-center problem

Problem Description

- network with node set N and arc set A: $G = (N, A)$
- set of demand nodes to be covered: $I \subset N$
- set of candidate facility sites: $J \subset N$
- maximum number of facilities: P
- distance of arc $(i, j) \in A$: d_{ij}

Goal: Locate a set of facilities that minimize the maximum distance between a demand node and its closest facility.

Decision variables

$$x_j = \begin{cases} 1, & \text{if we locate a facility at candidate site } j \in J, \\ 0, & \text{else} \end{cases}$$

$$y_{ij} = \begin{cases} 1, & \text{if customer node } i \in I \text{ is served by candidate site } j \in J, \\ 0, & \text{else} \end{cases}$$

$w =$ maximum over distances of all demand nodes and its closest facility

P-center problem

Decision variables

$$x_j = \begin{cases} 1, & \text{if we locate a facility at candidate site } j \in J, \\ 0, & \text{else} \end{cases}$$

$$y_{ij} = \begin{cases} 1, & \text{if customer node } i \in I \text{ is served by candidate site } j \in J, \\ 0, & \text{else} \end{cases}$$

$w =$ maximum over distances of all demand nodes and its closest facility

Model

minimize	w			minimize the maximum distance
subject to	$\sum_{j \in J} y_{ij}$	$= 1,$	$\forall i \in I,$	each demand node is covered by one facility
	$\sum_{j \in J} x_j$	$= P,$		number of facilities
	y_{ij}	$\leq x_j,$	$\forall i \in I,\ j \in J,$	each demand node can only be served by a located facilit
	$d_{ij} y_{ij}$	$\leq w,$	$\forall i \in I,\ j \in J,$	maximum distance
	x_j	$\in \{0, 1\},$	$\forall j \in J,$	binary variables
	y_{ij}	$\in \{0, 1\},$	$\forall i \in I,\ j \in J.$	

Discrete Location Models

- **Set covering:** min # sites needed to cover all demand nodes with C coverage distance
- **Max covering:** max # covered demands with P sites
- **P-center:** min # coverage distance needed to cover all demand nodes with P sites
- **P-median problem:** min total (or average) demand-weighted distance between demands and the nearest facility.

P-median problem

Problem Description

- network with node set N and arc set A: $G = (N, A)$
- set of demand nodes to be covered: $I \subset N$
- set of candidate facility sites: $J \subset N$
- maximum number of facilities: P
- distance of arc $(i, j) \in A$: d_{ij}
- demand at node i: h_i
- ~~cost to locate a facility at candidate site j: f_j for $j \in J$~~
- ~~coverage distance: C~~

Goal: Locate a set of facilities that minimizes the demand-weighted total (or average) distance between demands and the nearest facility.

P-median problem

Problem Description

- network with node set N and arc set A: $G = (N, A)$
- set of demand nodes to be covered: $I \subset N$
- set of candidate facility sites: $J \subset N$
- maximum number of facilities: P
- distance of arc $(i, j) \in A$: d_{ij}
- demand at node i: h_i

Goal: Locate a set of facilities that minimizes the demand-weighted total (or average) distance between demands and the nearest facility.

Decision variables

$$x_j = \begin{cases} 1, & \text{if we locate a facility at candidate site } j \in J, \\ 0, & \text{else} \end{cases}$$

$$y_{ij} = \begin{cases} 1, & \text{if customer node } i \in I \text{ is served by candidate site } j \in J, \\ 0, & \text{else} \end{cases}$$

P-median problem

Decision variables

$$x_j = \begin{cases} 1, & \text{if we locate a facility at candidate site } j \in J, \\ 0, & \text{else} \end{cases}$$

$$y_{ij} = \begin{cases} 1, & \text{if customer node } i \in I \text{ is served by candidate site } j \in J, \\ 0, & \text{else} \end{cases}$$

Model

$$\begin{array}{llll}
\text{minimize} & \displaystyle\sum_{i \in I, j \in J} h_i d_{ij} y_{ij} & & \text{minimize total demand-weighted distance} \\
\text{subject to} & \displaystyle\sum_{j \in J} y_{ij} = 1, & \forall i \in I, & \text{each demand node is served by one facility} \\
& \displaystyle\sum_{j \in J} x_j = P, & & \text{maximum number of facilities} \\
& y_{ij} \leq x_j, & \forall i \in I,\ j \in J, & \text{each demand node can only be served by a located facility} \\
& x_j \in \{0, 1\}, & \forall j \in J, & \\
& y_{ij} \in \{0, 1\}, & \forall i \in I,\ j \in J. &
\end{array}$$

P-center v.s. P-median problem

P-center problem:

$$
\begin{array}{lll}
\text{minimize} & w & \\
\text{subject to} & \sum_{j \in J} y_{ij} & = 1, \qquad \forall i \in I, \\
& \sum_{j \in J} x_j & = P, \\
& y_{ij} & \leq x_j, \qquad \forall i \in I,\ j \in J, \\
& d_{ij} y_{ij} & \leq w, \qquad \forall i \in I,\ j \in J, \\
& x_j & \in \{0,1\}, \quad \forall j \in J, \\
& y_{ij} & \in \{0,1\}, \quad \forall i \in I,\ j \in J.
\end{array}
$$

P-median problem:

$$
\begin{array}{lll}
\text{minimize} & \sum_{i \in I, j \in J} h_i d_{ij} y_{ij} & \\
\text{subject to} & \sum_{j \in J} y_{ij} = 1, & \forall i \in I, \\
& \sum_{j \in J} x_j = P, & \\
& y_{ij} \leq x_j, & \forall i \in I,\ j \in J, \\
& x_j \in \{0,1\}, & \forall j \in J, \\
& y_{ij} \in \{0,1\}, & \forall i \in I,\ j \in J.
\end{array}
$$

Discrete Location Models

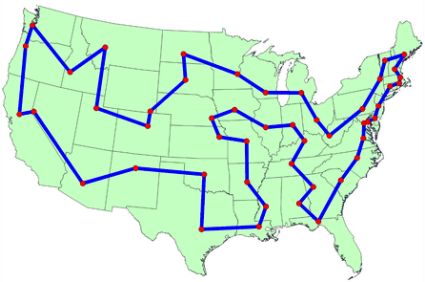
- **Set covering:** min # sites needed to cover all demand nodes with C coverage distance
- **Max covering:** max # covered demands with P sites
- **P-center:** min # coverage distance needed to cover all demand nodes with P sites
- **P-median problem:** min total (or average) demand-weighted distance between demands and the nearest facility.

Traveling Salesman Problem (TSP)

Traveling Salesman Problem (TSP)

A traveling salesman seeks to identify the tour that will allow him to

- start and end in the same city
- visit each city only once
- achieve the minimum cost.



In the 1950s, George Dantzig, Delber R. Fulkerson, and Selmer M. Johnson solved the 48 state problem by formulating it as an integer linear programming problem.

Source: `https://optimization.cbe.cornell.edu/index.php?title=Traveling_salesman_problem`

Recall: Traveling salesman problem

Consider a salesman who must visit six cities.

- Must start from city 1 and return to city 1.
- Must visit each city exactly once
- Every two cities i and j are connected with the distance d_{ij} for $i, j \in \{1, \ldots, 6\}$
- Formulate the problem as a mixed-integer linear programming problem to to minimize the total distance traveled.

Decision variables: binary variables x_{ij} denote whether to visit city j after visiting city i.

- $x_{ii} = 0$, $\sum_{j=1}^{6} x_{ij} = 1$, $\sum_{i=1}^{6} x_{ij} = 1$ for $i \in [6], j \in [6]$
- Subtour elimination: for all $S \subseteq \{2, 3, 4, 5, 6\}$, $\sum_{i \in S, j \in S} x_{i,j} \leq |S| - 1$.

Traveling Salesman Problem (TSP)

Problem description: For a graph $G = (N, A)$ where $N = \{1, \ldots, n\}$, a traveling salesman
- starts from node 1, and returns to node 1
- visit each node exactly once
- d_{ij} = distance from node i to node j

Denote for any $S \subseteq N$,

$$\begin{aligned}
\delta^-\{S\} &:= \{(i,j) \in A : i \notin S, j \in S\} && = \text{in-arcs of } S \\
\delta^+\{S\} &:= \{(i,j) \in A : i \in S, j \notin S\} && = \text{out-arcs of } S \\
A(S) &:= \{(i,j) \in A : i, j \in S\} && = \text{all arcs connecting nodes in } S
\end{aligned}$$

Goal: Find the route with the shortest total travel distance.

Decision variables

$$x_{ij} = \begin{cases} 1, & \text{if we visit node } i \text{ immediately followed by node } j, \\ 0, & \text{else} \end{cases}$$

Traveling Salesman Problem (TSP)

Decision variables

$$x_{ij} = \begin{cases} 1, & \text{if we visit node } i \text{ immediately followed by node } j, \\ 0, & \text{else} \end{cases}$$

Dantzig-Fulkerson-Johnson (DFJ) formulation [Dantzig et al. 1954]

Model 1

$$\begin{array}{llllll}
\text{minimize} & \displaystyle\sum_{(i,j)\in A} d_{ij} & x_{ij} & & & \text{minimize total travel distance} \\
\text{subject to} & \displaystyle\sum_{(j,i)\in\delta^-\{i\}} & x_{ji} & = 1, & \forall i \in N, & \text{exactly one arc enters node } i \\
 & \displaystyle\sum_{(i,j)\in\delta^+\{i\}} & x_{ij} & = 1, & \forall i \in N, & \text{exactly one arc leaves node } i \\
 & \displaystyle\sum_{(i,j)\in A(S)} & x_{ij} & \leq |S| - 1, & \forall 2 \leq |S| \leq n-1, & \text{subtour elimination constraints} \\
 & & x_{ij} & \in \{0,1\}, & \forall i,j \in N. & \text{binary variables}
\end{array}$$

Traveling Salesman Problem (TSP)

Dantzig-Fulkerson-Johnson (DFJ) formulation [Dantzig et al. 1954]

Model 2

minimize	$\sum_{(i,j)\in A} d_{ij} \quad x_{ij}$			minimize total travel distance
subject to	$\sum_{(j,i)\in\delta^-\{i\}} x_{ji}$	$= 1,$	$\forall i \in N,$	exactly one arc enters node i
	$\sum_{(i,j)\in\delta^+\{i\}} x_{ij}$	$= 1,$	$\forall i \in N,$	exactly one arc leaves node i
	$\sum_{(i,j)\in\delta^+\{S\}} x_{ij}$	$\geq 1,$	$\forall 2 \leq \lvert S\rvert \leq n-1,$	subtour elimination constraints
	x_{ij}	$\in \{0,1\},$	$\forall i,j \in N.$	binary variables

Traveling Salesman Problem (TSP)

Miller-Tucker-Zemlin (MTZ) formulation [Miller et al. 1960]

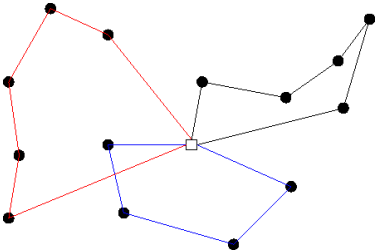
Additional decision variable: u_i = the position of node i in the visiting sequence.

minimize	$\sum_{(i,j)\in A} d_{ij}x_{ij}$			minimize total travel distance
subject to	$\sum_{(j,i)\in\delta^-\{i\}} x_{ji}$	$= 1,$	$\forall i \in N,$	exactly one arc enters node i
	$\sum_{(i,j)\in\delta^+\{i\}} x_{ij}$	$= 1,$	$\forall i \in N,$	exactly one arc leaves node i
	$u_i + 1 - n(1 - x_{ij})$	$\leq u_j,$	$\forall i, j \in N, j \neq 1,$	subtour elimination constraints
	$u_1 = 1,$			fixing the starting node
	$u_i \leq n,$		$\forall i \in N,$	bounding constraints
	$x_{ij} \in \{0, 1\},$		$\forall i, j \in N.$	binary variables

Vehicle Routing Problem (VRP)

Vehicle Routing Problem (VRP)

A homogeneous fleet of capacitated vehicles
- starts from the depot
- serves (pick-up or delivery) each of the n customers (with demand) exactly once
- returns to the depot



Goal: Find the route with the shortest total travel distance.

Source: `https://en.wikipedia.org/wiki/Vehicle_routing_problem`

Vehicle Routing Problem (VRP)

Problem description: For a graph $G = (N, A)$ where $N = \{0, 1, \ldots, n\}$, a homogeneous fleet of K capacitated vehicles

- starts from node 0, and returns to node 0
- pick up the packages at each of the n demand nodes exactly once

Moreover, we are given the following **data**:

q_i = demand at customer node i ($q_0 = 0$)
d_{ij} = distance from node i to node j
$\mathcal{K}$ = the index set of a fleet with $K = |\mathcal{K}|$
Q = vehicle capacity

Denote for any $S \subseteq N$,

$$\begin{aligned} \delta^-\{S\} &:= \{(i,j) \in A : i \notin S, j \in S\} && = \text{in-arcs of } S \\ \delta^+\{S\} &:= \{(i,j) \in A : i \in S, j \notin S\} && = \text{out-arcs of } S \\ A(S) &:= \{(i,j) \in A : i, j \in S\} && = \text{all arcs connecting nodes in } S \end{aligned}$$

Recall: MTZ formulation for TSP

Problem description: For a graph $G = (N, A)$ where $N = \{1, \ldots, n\}$, a traveling salesman
- starts from node 1, and returns to node 1
- visit each node exactly once
- d_{ij} = distance from node i to node j

minimize	$\sum_{(i,j)\in A} d_{ij}x_{ij}$			minimize total travel distance
subject to	$\sum_{(j,i)\in\delta^-\{i\}} x_{ji}$	$= 1,$	$\forall i \in N,$	exactly one arc enters node i
	$\sum_{(i,j)\in\delta^+\{i\}} x_{ij}$	$= 1,$	$\forall i \in N,$	exactly one arc leaves node i
	$u_i + 1 - n(1 - x_{ij})$	$\leq u_j,$	$\forall i, j \in N, j \neq 1,$	subtour elimination constraints
	$u_1 = 1,$			fixing the starting node
	$u_i \leq n,$		$\forall i \in N,$	bounding constraints
	$x_{ij} \in \{0, 1\},$		$\forall i, j \in N.$	binary variables

Vehicle Routing Problem (VRP)

Decision variables

$$x_{ij} = \begin{cases} 1, & \text{if a vehicle travels on arc } (i,j), \\ 0, & \text{else} \end{cases}$$

$u_i =$ (lower bound on the) vehicle load after visiting node i (pickup case)

Two-index vehicle flow model (from MTZ)

$$\begin{array}{llll}
\text{minimize} & \sum_{(i,j)\in A} d_{ij}x_{ij} & & \text{minimize total travel distance} \\
\text{subject to} & \sum_{(0,j)\in\delta^+\{0\}} x_{0j} & \leq K, & \text{fleet size constraint (at the depot)} \\
& \sum_{(j,i)\in\delta^-\{i\}} x_{ji} & = 1, \quad \forall i \in N \setminus \{0\}, & \text{exactly one arc enters node } i \\
& \sum_{(i,j)\in\delta^+\{i\}} x_{ij} & = 1, \quad \forall i \in N \setminus \{0\}, & \text{exactly one arc leaves node } i \\
& u_i + q_j - Q(1 - x_{ij}) & \leq u_j, \quad \forall i,j \in N \setminus \{0\}, & \text{subtour elimination} \\
& q_i \leq u_i & \leq Q, \quad \forall i \in N \setminus \{0\}, & \text{capacity constraints} \\
& x_{ij} & \in \{0,1\}, \quad \forall i,j \in N. & \text{binary variables}
\end{array}$$

Vehicle Routing Problem (VRP)

Two-index formulations is suitable for the case with homogeneous fleet.

To model vehicle-specific characteristics such as different capacities, associated depots, and costs, we need the three-index formulation, with additional index for the vehicle.

Decision variables

$$x_{ij}^k = \begin{cases} 1, & \text{if vehicle k travels on arc } (i,j), \\ 0, & \text{else} \end{cases}$$

$$u_i^k = \text{(lower bound on the) load of vehicle k after visiting node } i \text{ (pickup case)}$$

Vehicle Routing Problem (VRP)

Three-index vehicle flow model (from MTZ)

$$
\begin{array}{lll}
\underset{x_{ij}^k \in \{0,1\}, \forall i,j \in N, k \in \mathcal{K}}{\text{minimize}} & \displaystyle\sum_{k \in \mathcal{K}} \sum_{(i,j) \in A} d_{ij} x_{ij}^k & \\
\text{subject to} & \displaystyle\sum_{(0,j) \in \delta^+\{0\}} x_{0j}^k & \leq 1, \quad \forall k \in \mathcal{K}, \\
& \text{(each vehicle leaves the depot (or not))} & \\
& \displaystyle\sum_{(j,i) \in \delta^-\{i\}} x_{ji}^k - \sum_{(i,j) \in \delta^+\{i\}} x_{ij}^k & = 0, \quad \forall i \in N \setminus \{0\}, \forall k \in \mathcal{K}, \\
& \text{(each vehicle flow balance constraints)} & \\
& \displaystyle\sum_{(i,0) \in \delta^-\{0\}} x_{i0}^k & \leq 1, \quad \forall k \in \mathcal{K}, \\
& \text{(each vehicle returns to the depot (or not))} & \\
& \displaystyle\sum_{k \in \mathcal{K}} \sum_{(i,j) \in \delta^+\{i\}} x_{ij}^k & = 1, \quad \forall i \in N \setminus \{0\}, \\
& \text{(each customer is assigned to one route)} & \\
& u_i^k + q_j - Q(1 - x_{ij}^k) & \leq u_j^k, \quad \forall i,j \in N \setminus \{0\}, \forall k \in \mathcal{K}, \\
& \text{(subtour elimination)} & \\
& q_i \leq u_i^k & \leq Q, \quad \forall i \in N \setminus \{0\}
\end{array}
$$

Open Vehicle Routing Problem (OVRP)

Open Vehicle Routing Problem (OVRP)

Problem description: For a graph $G = (N, A)$ where $N = \{0, 1, \ldots, n\}$, a homogeneous fleet of K capacitated vehicles

- starts from node 0, and do not need to return to node 0
- pick up the packages at each of the n demand nodes exactly once

Moreover, we are given the following **data**:

q_i = demand at customer node i ($q_0 = 0$)
d_{ij} = distance from node i to node j
$\mathcal{K}$ = the index set of a fleet with $K = |\mathcal{K}|$
Q = vehicle capacity

Denote for any $S \subseteq N$,

$$\begin{aligned} \delta^-\{S\} &:= \{(i,j) \in A : i \notin S, j \in S\} && = \text{in-arcs of } S \\ \delta^+\{S\} &:= \{(i,j) \in A : i \in S, j \notin S\} && = \text{out-arcs of } S \\ A(S) &:= \{(i,j) \in A : i, j \in S\} && = \text{all arcs connecting nodes in } S \end{aligned}$$

Open Vehicle Routing Problem (OVRP)

Decision variables

$$x_{ij} = \begin{cases} 1, & \text{if a vehicle travels on arc } (i,j), \\ 0, & \text{else} \end{cases}$$

$u_i =$ (lower bound on the) vehicle load after visiting node i (pickup case)

Two-index vehicle flow model (from MTZ)

minimize	$\sum_{(i,j)\in A} d_{ij}x_{ij}$			minimize total travel distance
subject to	$\sum_{(0,j)\in\delta^+\{0\}} x_{0j}$	$\leq K,$		fleet size constraint (at the depot)
	$\sum_{(j,i)\in\delta^-\{i\}} x_{ji}$	$= 1,$	$\forall i \in N \setminus \{0\},$	exactly one arc enters node i
	$\sum_{(i,j)\in\delta^+\{i\}} x_{ij}$	$\leq 1,$	$\forall i \in N \setminus \{0\},$	at most arc leaves node i
	$u_i + q_j - Q(1 - x_{ij})$	$\leq u_j,$	$\forall i,j \in N \setminus \{0\},$	subtour elimination
	$q_i \leq u_i$	$\leq Q,$	$\forall i \in N \setminus \{0\},$	capacity constraints
	x_{ij}	$\in \{0,1\},$	$\forall i,j \in N.$	binary variables

Vehicle routing problem with time windows (VRPTW)

Vehicle routing problem with time windows (VRPTW)

Problem description: For a graph $G = (N, A)$ where $N = \{0, 1, \ldots, n\}$, a homogeneous fleet of K capacitated vehicles

- starts from node 0, and returns to node 0
- pick up the packages at each of the n demand nodes exactly once
- satisfies the time window requirement for each demand node

Moreover, we are given the following **data**:

q_i	= demand at customer node i ($q_0 = 0$)
d_{ij}	= distance from node i to node j
$\mathcal{K}$	= the index set of a fleet with $K = \lvert\mathcal{K}\rvert$
Q	= vehicle capacity
t_{ij}	= travel time from node i to node j
$[e_i, \ell_i]$	= time window to start service at node i
s_i	= service time at node i

Vehicle routing problem with time windows (VRPTW)

Decision variables

$$x_{ij} = \begin{cases} 1, & \text{if a vehicle travels on arc } (i,j), \\ 0, & \text{else} \end{cases}$$

$$u_i = \text{(lower bound on the) vehicle load after visiting } i$$

$$w_i = \text{time at which the service at node } i \text{ starts}$$

Vehicle routing problem with time windows (VRPTW)

Model: two-index vehicle flow model

$$
\begin{array}{llll}
\text{minimize} & \displaystyle\sum_{(i,j)\in A} d_{ij}x_{ij} & & \\
\text{subject to} & \displaystyle\sum_{(0,j)\in\delta^+\{0\}} x_{0j} & \leq K, & \\
& \displaystyle\sum_{(j,i)\in\delta^-\{i\}} x_{ji} & = 1, & \forall i \in N \setminus \{0\}, \\
& \displaystyle\sum_{(i,j)\in\delta^+\{i\}} x_{ij} & = 1, & \forall i \in N \setminus \{0\}, \\
& q_i \leq u_i & \leq Q, & \forall i \in N \setminus \{0\}, \\
& u_i + q_j - Q(1 - x_{ij}) & \leq u_j, & \forall i, j \in N \setminus \{0\}, \\
& \text{(subtour elimination)} & & \\
& w_i + s_i + t_{ij} - W_{ij}(1 - x_{ij}) & \leq w_j, & \forall i, j \in N \setminus \{0\}, \\
& \text{(time and subtour elimination)} & & \\
& e_i \leq w_i & \leq \ell_i, & \forall i \in N \setminus \{0\}, \\
& \text{(time window constraints)} & & \\
& x_{ij} & \in \{0, 1\}, & \forall i, j \in N.
\end{array}
$$

where $W_{i,j}$ should be selected satisfying $W_{i,j} \geq \max\{0, \ell_i + s_i + t_{ij} - e_j\}$.

Reading materials

What You Should Know About Location Modeling, Mark S. Daskin (2008)

What You Should Know about the Vehicle Routing Problem, Gilbert Laporte (2007)